ResearchShop Pipeline

Technical Internals Report

A complete technical reference for the gene extraction pipeline: architecture, data flow, module internals, benchmark results, configuration flags, and known issues.

Generated April 10, 2026 from source at `RS_SOFTWAREX/`

Pipeline stages covered: PubMed Search → Screening → Full Text → PubTator NER → Gemini Extraction → Gene Validation → CSV Output

Contents

1 Architecture Overview

1.1 System diagram

The pipeline spans two processes: an **Electron renderer** (TypeScript, UI) and a **Python subprocess** (spawned on each run). They communicate over stdout/stdin using a newline-delimited JSON protocol.

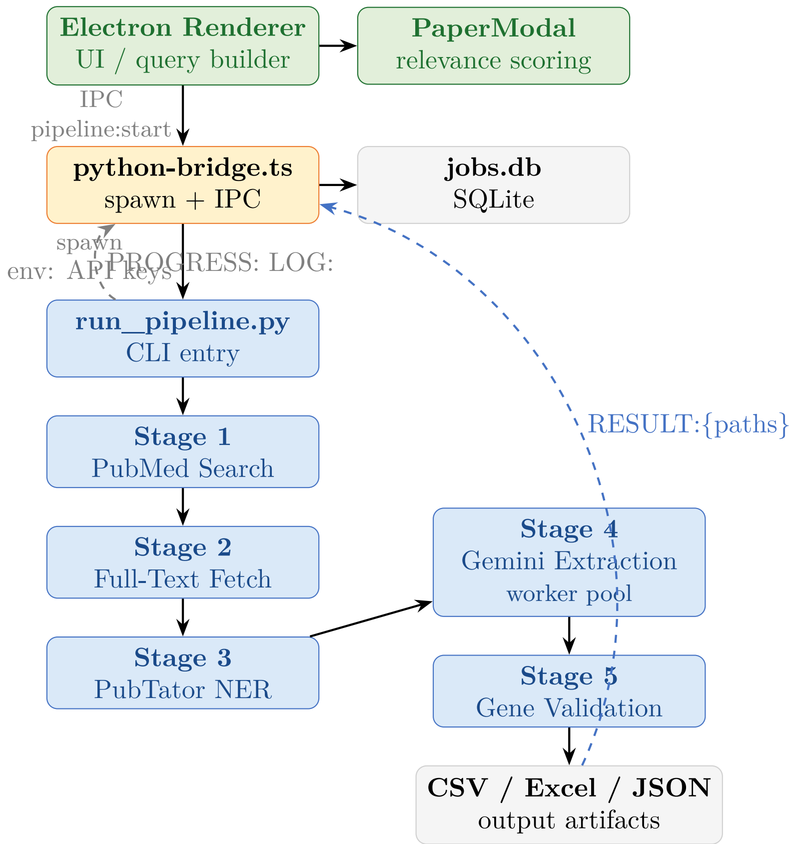

Figure 1: High-level system architecture. Dashed arrows show the stdout IPC protocol flowing back to the Electron main process.

1.2 The three trust tiers

The design resolves a fundamental tension: PubTator has high precision but low recall; Gemini has high recall but can hallucinate. The pipeline layers them:

Tier 1 — Gene Validator (HGNC + remote APIs): final safety gate, confidence scoring

Tier 2 — Gemini LLM (high recall / lower precision): grounding check + evidence gate

Tier 3 — PubTator NER (high precision / lower recall): precision floor

Each layer guards the one above it

1.3 The two process models

Mode	Config	Behaviour
Sequential (default)	`PARALLEL_ANALYSIS=false`	One paper at a time. Safe for free-tier Gemini (15 RPM
Parallel	`PARALLEL_ANALYSIS=true`	Multiple in-flight. For paid keys. Pool size: 2–4.

2 IPC Protocol

Python writes newline-delimited JSON to stdout. The Electron bridge parses it in a line buffer.

Prefix	Required fields	Optional / notes
`PROGRESS:`	`stage`, `percent`	`stats.gemini_api_calls`, `stats.genes_extracted`
`LOG:`	`level`, `msg`	`detail` (nullable)
`RESULT:`	*none enforced*	`local_path`, `metadata_path`, `excel_path`, `json_path`, `error`

[Bug #1.8] The `isResultPayload` type guard (`python-bridge.ts:26`) accepts any non-null object — including `{}`. A Python bug that emits `RESULT:{}` marks the job *completed* with null artifact paths.

Listing 1: `run_pipeline.py:46-52` — Emit protocol with mandatory flush

```python
def progress_callback(stage, pct, stats):
    msg = json.dumps({"stage": stage, "percent": pct, "stats": stats})
    print(f"PROGRESS:{msg}", flush=True)    # flush=True is mandatory

def log_callback(level, msg, detail=None):
    payload = json.dumps({"level": level, "msg": msg, "detail": detail})
    print(f"LOG:{payload}", flush=True)
```

Secrets are passed only via environment variables (`GEMINI_API_KEY`, `ENTREZ_EMAIL`). CLI args are visible in `ps aux` output and are never used for secrets.

3 Pipeline Stages

3.1 Stage 1 — PubMed Search & Citation Ranking

`python/modules/pubmed_data_collector.py` (457 lines)

Queries NCBI Entrez, applies the OA filter (`"loattrfull text[sb]"`), and ranks results by citation count.

Citation ranking chain: iCite (NIH, batch) → Semantic Scholar (fallback, per-PMID with 200 ms sleep). If both fail, results are unranked.

Overfetch factor: `ANALYSIS_OVERFETCH_FACTOR=4`. If the user requests top-10 papers, 40 candidates are analysed. Compensates for paywalled papers and low-extraction failures.

[Suspect #2.4] Semantic Scholar fallback is per-PMID with 200 ms sleep. For 1000 unresolved PMIDs: >3 minutes of pure waiting. Batch API exists but is not used.

3.2 Stage 2 — Gene Relevance Scoring (UI-side)

`src/renderer/utils/geneRelevanceScorer.ts`

Since 2026-03-02, relevance scoring moved from the Python pipeline to the Electron renderer. Users see tier badges (**high** $\geq$10, **medium** $\geq$5, **low** 1–4, **none** $\leq$0) and can override the filter. The pipeline receives only the user's final PMID selection — no silent post-submission filtering.

The Python `abstract_screener.py` is retained for benchmarking and writes forensic `ScreeningDecision` records to the debug artifact, but no longer gates papers.

3.3 Stage 3 — Full-Text Fetch

`python/modules/full_text_fetcher.py` (1036 lines)

Strategy chain: PMC JATS XML → Europe PMC → abstract-only.

Section keys produced: `abstract`, `introduction`, `methods`, `results`, `discussion`, `conclusion`, `supplementary`, `_preamble`.

Figures are downloaded (max 3 per paper, max 5 MB each) and passed as image bytes to the Gemini multimodal stage. Supplementary CSV/Excel/ZIP files are parsed separately.

Greek letter transliteration (W1 fix): α →alpha, β →beta etc. prevents gene variant strings like β-globin being corrupted.

3.4 Stage 4 — PubTator NER

`python/modules/pubtator_tool.py` (620 lines)

Submits PMIDs to the NCBI PubTator3 BioC API in batches of 10. Returns high-precision gene symbols as the *precision floor* for the hybrid architecture. Parse errors on individual papers in a batch are silently skipped (W10 — accepted known limitation).

3.5 Stage 5 — Gemini Extraction

`python/modules/gemini_extractor.py` (2230 lines)

The most complex module. See Section 4 for full detail.

3.6 Stage 6 — Gene Validation

`python/modules/gene_validator.py` (1122 lines)

Source chain: Local HGNC JSON (44,933 genes) → HGNC REST API → MyGene.info → fuzzy match.

Returns `validation_confidence` (0.0–1.0) and `validation_source`. The confidence threshold `FINAL_VALIDATION_MIN_CONFIDENCE=0.7` is a medical accuracy decision — do not lower without benchmark evaluation.

Citation validation uses dense sequence matching (`difflib.SequenceMatcher`) with a 0.85 ratio threshold and gene context window of ±1500 characters.

3.7 Stage 7 — Orchestration & CSV Output

`python/modules/pipeline_orchestrator.py` (1686 lines)

Manages the worker pool, aggregates results, computes confidence tiers, deduplicates, and writes four output artifacts: primary CSV, metadata CSV, Excel workbook (two sheets), JSON.

4 Gemini Extraction Deep Dive

4.1 Class lifecycle and mutable state

`GeneInfoPipeline` is instantiated *once per paper* inside the worker process. State never leaks across papers.

Attribute	Type	Lifecycle
`paper_text`	`str`	Set in `__init__`; may be truncated by context validation
`original_paper_text`	`str`	Set in `__init__`; immutable backup
`associations`	`List[Dict]`	Reset in step 0; mutated by each discovery stage
`candidate_meta`	`Dict[Tuple, Dict]`	Reset in step 0; accumulates provenance for all candidates
`dropped_candidates`	`List[Dict]`	Reset in step 0; audit trail for dropped genes
`strict_gate_drops`	`List[Dict]`	Reset in step 0; confidence gate drops
`_paper_api_calls`	`int`	Incremented on each LLM call; returned to orchestrator

4.2 The `run_pipeline()` orchestrator

After the 2026-04-07 readability refactor, the main method is a 25-line orchestrator delegating to five named sub-stages:

Listing 2: `gemini_extractor.py:1318-1356` — post-refactor orchestrator

```python
def run_pipeline(self, column_descriptions):
    # Step 0: Validate and prepare paper text for context windows
    context_validation = self._validate_and_prepare_paper_text()
    if context_validation["failed"]:
        return pd.DataFrame()

    # Steps 0.5-1.5: Candidate discovery
    self._run_candidate_discovery()

    # Step 1.6: Grounding check
    self._run_grounding_check()

    # Step 2: Gene validation + normalization
    self._run_validation_and_normalize()

    # Step 3: Detail extraction
    extracted_info = self._run_detail_extraction(column_descriptions)
    if not extracted_info:
        return pd.DataFrame()

    # Step 4: Post-validation (metadata, strict gate, citation, evidence gate)
    df = pd.DataFrame(extracted_info)
    if "variant_name" in df.columns:
        df["variant_name"] = df["variant_name"].apply(
            self._normalize_variant_value)
    return self._run_post_validation(df, column_descriptions,
                                     context_validation)
```

4.3 The five sub-stages

`_run_candidate_discovery()` Steps 0.5–1.5. Resets all mutable state, then runs: abstract pass (LLM), full-text pass ×2 (temperature 0.0 then 0.4), HGNC deterministic lexicon, figure multimodal analysis, PubTator merge. Each source contributes to `candidate_meta` with a source label (`llm_text`, `llm_abstract`, `pubtator`, `deterministic_lexicon`, `llm_figure`).

`_run_grounding_check()` Step 1.6. The primary hallucination filter. Each candidate is searched for in the paper text using its canonical symbol, HGNC aliases, and raw LLM labels (e.g. "BNP" for NPPB). Figure-only genes use a lighter caption-text check. Ungrounded candidates are dropped and marked `rejected_ungrounded`.

`_run_validation_and_normalize()` Step 2. Calls `_apply_gene_validation_heuristics()` (HGNC chain), then ensures one gene-level row (variant = "") per gene. If strict gate is off and all associations are filtered, falls back to pre-validation list.

`_run_detail_extraction(column_descriptions)` Step 3. All validated candidates go into ONE Gemini call with the user's column schema. Returns merged, backfilled rows, or fallback rows if the LLM produced nothing.

`_run_post_validation(df, ...)` Step 4. Adds metadata, applies strict validation gate (drops < 0.7 confidence), citation validation (adds `*_citation_valid` booleans), evidence gate (per-source evidence thresholds), and context window metadata.

4.4 The four LLM extraction methods

Method	Input	Max retries	Backoff
`extract_gene_names_from_abstract`	Abstract only	2	3 s flat
`extract_gene_names` (×2)	Full text	3	5s ×2 exp
`extract_gene_names_from_figures`	Image + caption	3/figure	Retry-after header
`extract_gene_info` (Stage 3)	Full text + all genes	3	5s ×2 exp

All calls use `thinking_budget=0` (C20 fix — preview models hang >600 s without this).

4.5 Confidence tiers

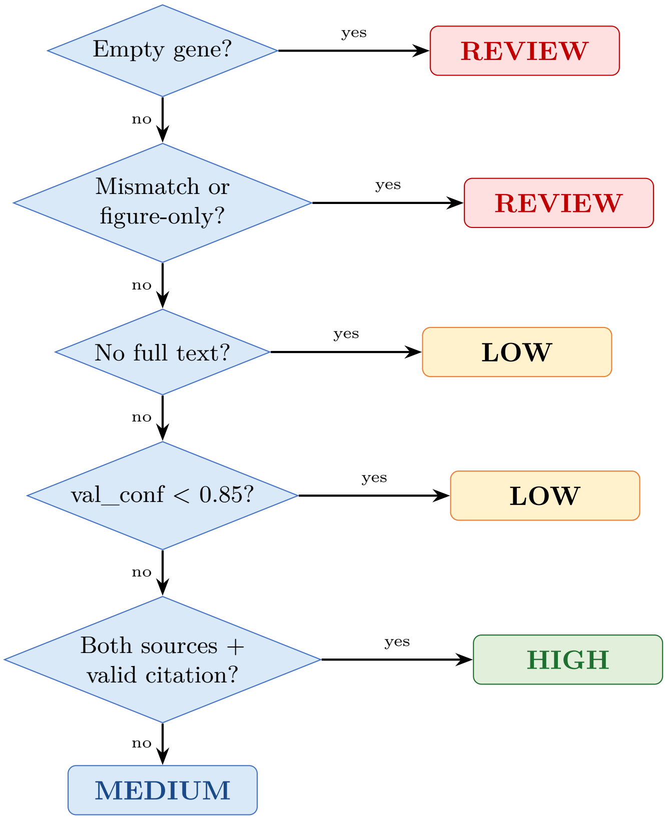

Figure 2: Confidence tier decision tree (`pipeline_orchestrator.py:27-101`).

[Suspect #2.1] HIGH is unreachable when `val_conf` < 0.85 even with dual-source corroboration. A gene confirmed by both PubTator and Gemini but with a borderline 0.80 HGNC confidence lands in LOW rather than MEDIUM.

5 Benchmark Results

Evaluated on 12 papers across five study types, three runs each (36 total runs). Open-access full-text papers only.

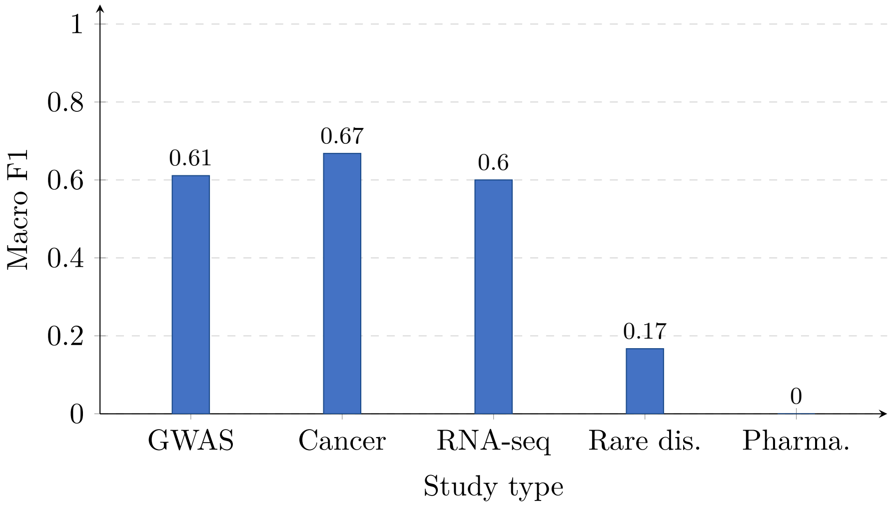

Figure 3: Macro F1 by study type. Rare disease is low because one of the two papers (PMID 21076407) is paywalled, returning 0 genes. Pharmacogenomics F1=0 reflects that guidelines embed gene-drug associations in dosing tables, not primary research findings — a design limitation, not a bug.

PMID	Type	P	R	F1	Notes
17463248	GWAS	1.000	1.000	1.000	T2D — perfect extraction
21926974	GWAS	1.000	0.714	0.833	Schizophrenia
17554300	GWAS	0.000	0.000	0.000	Crohn's — OA text issue
21720365	Cancer genomics	0.875	1.000	0.933	TCGA ovarian
23000897	Cancer genomics	0.667	0.667	0.667	TCGA breast
24132290	Cancer genomics	0.254	1.000	0.405	Pan-cancer; 59 genes vs 15 gold
20129251	RNA-seq	1.000	1.000	1.000	GBM — figure analysis critical
32416070	RNA-seq	1.000	0.111	0.200	Only 1/9 gold genes extracted
19915526	Rare disease	0.200	1.000	0.333	Miller syndrome
21076407	Rare disease	0.000	0.000	0.000	Paywalled — abstract only
19228618	Pharmacogenomics	0.000	0.000	0.000	Guideline document
22205192	Pharmacogenomics	0.000	0.000	0.000	Guideline document

Repeatability: Jaccard similarity = 1.0 on all 12 papers across 3 runs. The gene set is perfectly reproducible run-to-run (deterministic candidate seeding).

Figure analysis: Controlled 36-run experiment. PMID 20129251 (GBM): F1-on = 1.000, F1-off = 0.167 (ΔF1 = +0.833). Figures are the primary communication medium in comprehensive genomics papers; the LLM uses them as editorial anchors that improve precision.

6 Configuration Flags

All flags live in `python/modules/config.py`. All boolean flags accept `"true"`/`"false"` env vars.

Flag	Default	Purpose	Stage
`ENABLE_OA_FILTER`	True	Restrict to open-access papers	1
`ENABLE_PUBTATOR_EXTRACTION`	True	Run NER precision floor	4

Continued on next page

Table 1 — continued

Flag	Default	Purpose	Stage
`ENABLE_FIGURE_ANALYSIS`	True	Multimodal figure extraction	5
`FIGURE_MAX_IMAGES_PER_PAPER`	3	Cap on figures per paper	5
`ENABLE_ABSTRACT_GENE_DISCOVERY`	True	Independent abstract-level LLM pass	5
`ENABLE_GROUNDING_CHECK`	True	Drop genes not found in text	5
`ENABLE_DETERMINISTIC_CANDIDATES`	True	HGNC symbol regex extraction	5
`DETERMINISTIC_MAX_CANDIDATES`	120	Cap on deterministic candidates	5
`ENABLE_BIOMARKER_NORMALIZATION`	True	Resolve BNP→NPPB etc.	5
`ENABLE_STRICT_VALIDATION_GATE`	True	Drop rows below confidence threshold	5
`FINAL_VALIDATION_MIN_CONFIDENCE`	0.7	**Medical accuracy threshold**	5
`ENABLE_EVIDENCE_BACKFILL`	True	Auto-fill empty Key Finding fields	5
`ENABLE_STRICT_EVIDENCE_GATE`	True	Drop rows with insufficient evidence	5
`EVIDENCE_MIN_NONEMPTY_CELLS_LLM_TEXT`	0	LLM rows trusted without backfill	5
`EVIDENCE_MIN_NONEMPTY_CELLS_DETERMINISTIC`	1	Deterministic rows need corroboration	5
`ENABLE_CITATION_VALIDATION`	True	Validate citations exist in paper text	6
`CITATION_MIN_CONFIDENCE`	0.7	Citation validation threshold	6
`AI_PER_PAPER_TIMEOUT_SECONDS`	600	Worker timeout (10 min)	7
`AI_WORKER_POOL_SIZE`	2	Pool size, hard cap 4	7
`PARALLEL_ANALYSIS`	False	In-flight parallel mode	7
`ANALYSIS_OVERFETCH_FACTOR`	4	Fetch 4× requested top-N	7
`CONTEXT_SAFETY_MARGIN`	0.8	Use 80% of model context limit	5
`GEMINI_FLASH_CONTEXT_LIMIT`	1M	Flash model token limit	5

Note: `FINAL_VALIDATION_MIN_CONFIDENCE=0.7` is a medical accuracy decision. Lowering it increases false gene associations in the output. Do not change without full benchmark evaluation.

7 Known Issues & Bugs

7.1 Critical findings

Location	Issue	Suggested fix
`orchestrator.py: 1085-1092`	Worker `.get(timeout=0)` bare `except` swallows all error types. Segfaults, OOM kills, assertion errors all appear as generic `"error": str(e)`. No traceback preserved.	Log `traceback.format_exc()` and exception type before wrapping.
`orchestrator.py: 1169-1176`	Pool restart resets `submitted_at` to `time.time()`. A paper that hangs twice runs for up to $2 \times 600\,\mathrm{s} = 20\,\mathrm{min}$ total before the second timeout fires.	Track `original_submitted_at` that never resets; cap at 1 restart.
`orchestrator.py: 1563-1565`	`_agg_variants` silently returns `""` for empty-variant series. Gene-level rows lose the "variant = " signal post-dedup, potentially shifting confidence tier.	Emit `"(gene-level)"` sentinel or track gene-level evidence separately.
`orchestrator.py: 1573-1574`	Entire dedup step wrapped in swallow-all `except`. Duplicates silently appear in output.	Validate dtypes before groupby; emit LOG:error on skip.
`gene_validator.py: 704`	Citation dense-match threshold 0.85 is hardcoded. Paraphrased citations (synonyms, passive voice) never reach this ratio, causing valid rows to land in REVIEW.	Add `CITATION_MATCH_RATIO_THRESHOLD` to config.py.
`python-bridge.ts: 25-27`	`isResultPayload` accepts any non-null object. `RESULT:{}` marks job *completed* with all artifact paths null.	Require at least one of `local_path | error` to be a string.
`gemini_extractor.py 1455-1479`	Grounding check bypass: genes with `sources = {llm_figure, deterministic_lexicon}` skip the figure-caption check and may pass on weak lexical evidence.	Apply figure-caption check as mandatory AND-gate for any gene with `llm_figure` in sources.

7.2 Suspect / Fragile findings

Location	Issue	Severity
`orchestrator.py: 79-101`	HIGH tier unreachable when `val_conf` < 0.85, even with dual-source corroboration.	Suspect
`gene_validator.py: 216-250`	HGNC API fallback has no circuit breaker. 2000 genes $\times$ 8 s timeout = 4.5 hours on outage.	Suspect
`full_text_fetcher.py 354-413`	Multi-panel figures (1A, 1B, 1C) sharing a URL are deduped to one entry.	Fragile
`pubmed_data_ collector.py`	Semantic Scholar per-PMID fallback: 1000 unresolved PMIDs = 3+ minutes.	Suspect
`python-bridge.ts: 234-239`	RESULT vs. cancel micro-race window (microseconds; theoretical in single-threaded Node.js).	Fragile
`orchestrator.py: 409-517`	`_finalize_paper_result` silently omits `Gene Source` / `NCBI Gene ID` columns when PubTator is disabled.	Fragile

Table 3 — continued

Location	Issue	Severity
`orchestrator.py: 217-290`	`_write_split_output` silently drops user columns if renamed upstream. Schema mismatch between primary and metadata CSVs.	Fragile

7.3 Hardcoded constants to move to config

File:line	Value	Proposed flag
`gene_validator.py:704`	0.85 ratio	`CITATION_MATCH_RATIO_THRESHOLD`
`gene_validator.py:690`	0.6 word overlap	`CITATION_WORD_OVERLAP_THRESHOLD`
`gemini_extractor.py`	240 char snippet	already `EVIDENCE_SNIPPET_MAX_CHARS` ✓
`gemini_extractor.py`	20 PubTator genes in prompt	`PUBTATOR_PROMPT_MAX_GENES`
`orchestrator.py:79,88`	0.85 LOW tier threshold	`CONFIDENCE_THRESHOLD_FOR_LOW`
`full_text_fetcher.py`	200 rows CSV limit	`SUPPLEMENTARY_CSV_MAX_ROWS`
`pubmed_data_collector.py`	0.4 s fetch sleep	`PUBMED_FETCH_SLEEP_SECONDS`

8 Data Flow Between Stages

Hop	Key fields	Contract
Stage 1 → UI	pmid, title, abstract, citation_count	Missing pmid silently drops pape
UI → Python	JSON array of PMIDs (CLI arg)	Must be digit strings
Stage 3 → Stage 4	content, abstract, sections, figures, tables	`content` key must exist
Stage 4 → Stage 5	text, cols, pubtator_genes, figure_inputs, abstract	All must be picklable
Stage 5 → Stage 7	records, debug, gemini_api_calls	OR `error` string
Stage 7 → Bridge	local_path, metadata_path, excel_path, json_path	OR `error` string

Multiprocessing isolation: each worker process gets its own `GeneInfoPipeline` instance. Data crosses the process boundary via pickle. Image bytes in `figure_inputs` can be >5 MB per paper on multi-panel figures.

9 Prompt Engineering Notes

Four module-level constants in `gemini_extractor.py:25-86`:

`_GENE_DISCOVERY_INSTRUCTION_ABSTRACT` Abstract-level. Focuses on HUMAN genes, excludes model organisms, includes non-coding RNA only if primary finding. Contains the clinical-vs-molecular disambiguation clause.

`_GENE_DISCOVERY_INSTRUCTION_FULLTEXT` Full-text. Same as abstract version plus growth factors, receptors.

`_FIGURE_ANALYSIS_INSTRUCTION` Short. Key constraint: "Do not guess genes that are not explicitly shown."

`_DETAIL_EXTRACTION_CRITICAL_INSTRUCTIONS` 14 accumulated rules for Stage 3. Each rule addresses a specific observed failure mode (see `AUDIT.md` C8–C22).

Why no static clinical blocklist. A blocklist for ESR/AST/CRP blocks valid extractions in other paper types (ESR1 in breast cancer, GOT1/AST in liver genetics). A prompt disambiguation clause reads sentence context instead. The corroboration gate provides a hard backstop when the LLM fails the disambiguation.

Thinking mode disabled. `thinking_budget=0` on all `GenerateContentConfig` calls. Without it, Gemini preview models hang >600 s on 12k-token Stage 3 prompts (C20).

10 Suggested Triage Order

1. **Hardcoded citation thresholds** (§7, items 1–2) — easy config add, unblocks benchmarking tuning.
2. **Variant dedup silent loss** (`orchestrator.py:1563-1565`) — small fix, affects confidence tiers.
3. **Pool restart bugs** (`orchestrator.py:1169-1176`) — fix before parallel mode ships widely.
4. **Worker error swallowing** (`orchestrator.py:1085-1092`) — improves observability for everything else.
5. **Bridge trust boundary** (`python-bridge.ts:25-27`) — `isResultPayload` too permissive.
6. **Grounding bypass for mixed sources** — figure-only check should be an AND gate.

References

- Source code: `RS_SOFTWAREX/` repository
- Deep reference: `docs/pipeline/internals.md`
- Bug-hunting cheatsheet: `docs/pipeline/bug-hunting.md`
- Historical bug log: `AUDIT.md`
- Benchmark data: `python/data/benchmark/benchmark_results.csv`
- Project routing: `CLAUDE.md`